An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

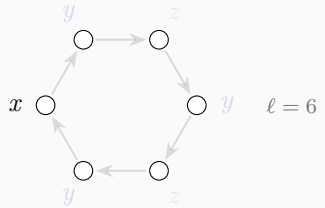
Yuya Uezato Yoshiki Nakamura

CyberAgent, Inc. Chiba University

STOC 2027 submission (author video for the PC)

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$\ell = 6$

$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

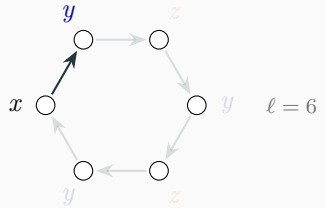
スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi| n^k)$。

メモ：

- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
- x を固定し y, z を交互に使えば 3 変数で足りる

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☞ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

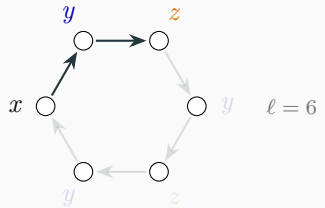
スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi| n^k)$。

メモ：

- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
- x を固定し y, z を交互に使えば 3 変数で足りる

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$\ell = 6$

$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☞ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

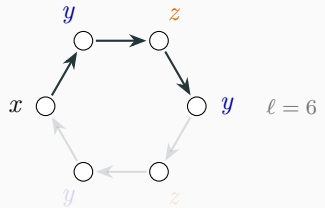
スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi| n^k)$。

メモ：

- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
- x を固定し y, z を交互に使えば 3 変数で足りる

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

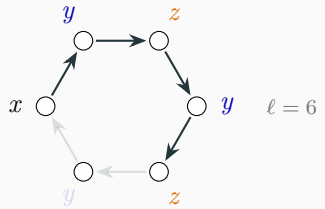
スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi| n^k)$。

メモ：

- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
- x を固定し y, z を交互に使えば 3 変数で足りる

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☞ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

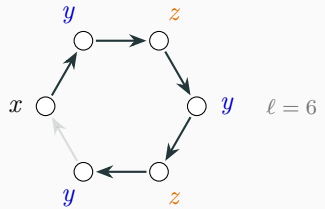
スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi| n^k)$。

メモ：

- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
- x を固定し y, z を交互に使えば 3 変数で足りる

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☞ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

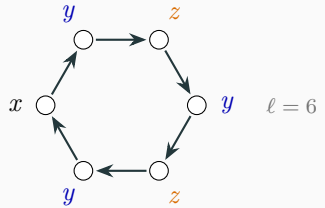
スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi| n^k)$。

メモ：

- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
- x を固定し y, z を交互に使えば 3 変数で足りる

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

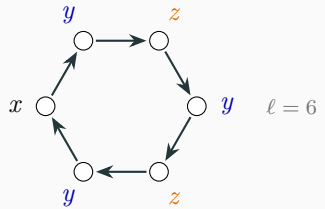
スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi| n^k)$。

メモ：

- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
- x を固定し y, z を交互に使えば 3 変数で足りる

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$$\exists y\left(E(x,y)\wedge\exists z\left(E(y,z)\wedge\exists y\left(E(z,y)\wedge\exists z\left(E(y,z)\wedge\exists y\left(E(z,y)\wedge E(y,x)\right)\right)\right)\right)\right)$$

Model checking: $\mathcal{M}\models\varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a,b\in\mathcal{M}$)

☛ $O(|\varphi|\cdot n^k)$ for FO^k by table memoization [Var95, PODS]

スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi|n^k)$。

メモ：

- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
- x を固定し y,z を交互に使えば 3 変数で足りる

FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z.\ - \wedge -$, we use DNF translation.
$\rightsquigarrow$ $2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

スライド概要：$k \geq 4$ では n^k が最適、FO^3 だけ n^ω に下がる。代償は翻訳の $2^{O(|\varphi|)}$。

メモ：

- 合成 = ブール行列積
- $\exists$ の内側の DNF 化が指数爆発の原因

FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z. - \wedge -$, we use DNF translation.
$\rightsquigarrow$ $2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

スライド概要：$k \geq 4$ では n^k が最適、FO^3 だけ n^ω に下がる。代償は翻訳の $2^{O(|\varphi|)}$。

メモ：

- 合成 = ブール行列積
- $\exists$ の内側の DNF 化が指数爆発の原因

FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z.\ - \wedge -$, we use DNF translation.
$\rightsquigarrow$ $2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

スライド概要：$k \geq 4$ では n^k が最適、FO^3 だけ n^ω に下がる。代償は翻訳の $2^{O(|\varphi|)}$。

メモ：

- 合成 = ブール行列積
- $\exists$ の内側の DNF 化が指数爆発の原因

FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z. - \wedge -$, we use DNF translation.
$\rightsquigarrow$ $2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

スライド概要：$k \geq 4$ では n^k が最適、FO^3 だけ n^ω に下がる。代償は翻訳の $2^{O(|\varphi|)}$。

メモ：

- 合成 = ブール行列積
- $\exists$ の内側の DNF 化が指数爆発の原因

Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\begin{aligned} & \exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big) \\ \rightsquigarrow_{\text{(DNF)}}\ & \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big) \\ \rightsquigarrow_{\text{(to CoR)}}\ & \Big(\bigcup_{A \subseteq [k]} \Big(\bigcap_{i \in A} P_i \Big) ; \Big(\bigcap_{i \notin A} Q_i \Big) \Big)(x,y) \end{aligned}$$

(In this translation, we use 2^k compositions)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS]**)**

Does every translation from FO^3 to CoR need $2^{\Omega(|\varphi|)}$ compositions in the worst case?

Partial answer: Yes for $\Sigma_1\mathrm{CoR}$ (with transitive closure) [Nak20, RAMiCS; Nak22, JLAMP]

The argument of [Nak22, JLAMP] is based on the fact that $\Sigma_1\mathrm{CoR}$ is equivalent to **existential** FO^3, which is closed under **extensions**.

スライド概要：この指数個の合成は必要か？ 未解決だった（$\Sigma_1\mathrm{CoR}$ のみ既知）。

メモ：

- 例の式は素直に訳すと合成 2^k 個
- 今回はサイズの無条件下界

Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big)$$

$$\rightsquigarrow_{\text{(DNF)}}\ \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big)$$

$$\rightsquigarrow_{\text{(to CoR)}}\ \Big(\bigcup_{A \subseteq [k]} \big(\bigcap_{i \in A} P_i \big) ; \big(\bigcap_{i \notin A} Q_i \big) \Big)(x,y)$$

(In this translation, we use 2^k compositions)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Does every translation from FO^3 to CoR need $2^{\Omega(|\varphi|)}$ compositions in the worst case?

Partial answer: Yes for $\Sigma_1\mathrm{CoR}$ (with transitive closure) [Nak20, RAMiCS; Nak22, JLAMP]
The argument of [Nak22, JLAMP] is based on the fact that $\Sigma_1\mathrm{CoR}$ is equivalent to **existential** FO^3, which is closed under **extensions**.

スライド概要：この指数個の合成は必要か？ 未解決だった（$\Sigma_1\mathrm{CoR}$ のみ既知）。

メモ：

- 例の式は素直に訳すと合成 2^k 個
- 今回はサイズの無条件下界

Outline

1. Our results

2. Technical overview

3. Conclusion

Our results

The answer is **YES!**

Thm. (This paper)

For each k, there is an FO^3 formula of size $O(k)$ such that every equivalent CoR term needs $\geq 2^{k-1}$ compositions, even with complement, converse, and shared subterms.

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ before: exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
  $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with transitive closure X^*

👉 Our proof technique can apply to all the above!

スライド概要：答えは YES（主定理）。同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界。

メモ：

- FO^2：$O(k)$ サイズの FO^3 文、同値な FO^2 文は量化子 $\geq 2^k$
- 行列：集約ゲート $\geq 2^k/(3\beta)$（β：半環の join breadth）
- 閉包付き CoR：$2 \cdot \#\text{合成} + \#\text{閉包} \geq 2^k$
- 行列言語：Brijder–Gyssens–Van den Bussche の問いに答える
- 変わるのは「1 演算が気づける変更の数」だけ（量化子 1、集約 3β、閉包 1）
- 論文には他に：CoR 等式、逆なし回路（ちょうど 2^k）、乱択回路、ランダム入力

Our results

The answer is **YES!**

Thm. (This paper)

For each k, there is an FO^3 formula of size $O(k)$ such that every equivalent CoR term needs $\geq 2^{k-1}$ compositions, even with complement, converse, and shared subterms.

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ before: exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
  $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with transitive closure X^*

👉 Our proof technique can apply to all the above!

スライド概要：答えは YES（主定理）。同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界。

メモ：

- FO^2：$O(k)$ サイズの FO^3 文、同値な FO^2 文は量化子 $\geq 2^k$
- 行列：集約ゲート $\geq 2^k/(3\beta)$（β：半環の join breadth）
- 閉包付き CoR：$2 \cdot \#\text{合成} + \#\text{閉包} \geq 2^k$
- 行列言語：Brijder–Gyssens–Van den Bussche の問いに答える
- 変わるのは「1 演算が気づける変更の数」だけ（量化子 1、集約 3β、閉包 1）
- 論文には他に：CoR 等式、逆なし回路（ちょうど 2^k）、乱択回路、ランダム入力

Our results

The answer is **YES!**

Thm. (This paper)

For each k, there is an FO^3 formula of size $O(k)$ such that every equivalent CoR term needs $\geq 2^{k-1}$ compositions, even with complement, converse, and shared subterms.

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ before: exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
  $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with transitive closure X^*

👉 Our proof technique can apply to all the above!

スライド概要：答えは YES（主定理）。同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界。

メモ：

- FO^2：$O(k)$ サイズの FO^3 文、同値な FO^2 文は量化子 $\geq 2^k$
- 行列：集約ゲート $\geq 2^k/(3\beta)$（β：半環の join breadth）
- 閉包付き CoR：$2 \cdot \#\text{合成} + \#\text{閉包} \geq 2^k$
- 行列言語：Brijder–Gyssens–Van den Bussche の問いに答える
- 変わるのは「1 演算が気づける変更の数」だけ（量化子 1、集約 3β、閉包 1）
- 論文には他に：CoR 等式、逆なし回路（ちょうど 2^k）、乱択回路、ランダム入力

Our results

The answer is **YES!**

Thm. (This paper)

For each k, there is an FO^3 formula of size $O(k)$ such that every equivalent CoR term needs $\geq 2^{k-1}$ compositions, even with complement, converse, and shared subterms.

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ before: exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
  $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with transitive closure X^*

👉 Our proof technique can apply to all the above!

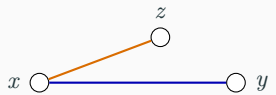
スライド概要：答えは YES（主定理）。同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界。

メモ：

- FO^2：$O(k)$ サイズの FO^3 文、同値な FO^2 文は量化子 $\geq 2^k$
- 行列：集約ゲート $\geq 2^k/(3\beta)$（β：半環の join breadth）
- 閉包付き CoR：$2 \cdot \#\text{合成} + \#\text{閉包} \geq 2^k$
- 行列言語：Brijder–Gyssens–Van den Bussche の問いに答える
- 変わるのは「1 演算が気づける変更の数」だけ（量化子 1、集約 3β、閉包 1）
- 論文には他に：CoR 等式、逆なし回路（ちょうど 2^k）、乱択回路、ランダム入力

The hard formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \big[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z))\big] \qquad (|G_k| = O(k))$$



at every i: one of the edges xy, xz is in P_i,
the other is in Q_i

DNF: $G_k \equiv \bigcup_{A \subseteq [k]} \big[H_A \cap (H_{A^c}\,; \mathbf{1})\big]$: 2^k **compositions** $(H_A \coloneqq \bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i)$

$\Rightarrow$ optimal up to a factor of 2: every equivalent CoR term needs $\geq 2^{k-1}$ compositions

スライド概要：G_k は DNF で合成 2^k 個、これは 2 倍を除いて最適。

メモ：

- 各 i で、辺 xy と辺 xz の一方が P_i、他方が Q_i
- 下界は合成以外の演算（補・逆・共有）を無料にしても成り立つ

Outline

1. Our results

2. Technical overview

3. Conclusion

Proof idea: telling $\mathcal{M}_0$ apart from 2^k structures

Goal $C \equiv G_k \implies C$ has $\geq 2^{k-1}$ compositions (C: any CoR term or circuit)

C must tell $\mathcal{M}_0$ apart from each of 2^k structures $\mathcal{M}_\lambda$: $G_k = 1$ on $\mathcal{M}_0$, $G_k = 0$ on $\mathcal{M}_\lambda$

$\mathcal{M}_0$

CoR operation	meaning	newly tells $\mathcal{M}_0$ apart from
$\mathbf{0}, \mathbf{1}, \mathbf{id}$, $\cup, \cap, \neg$, $X^\smile$	constants, Boolean operations, converse	no $\mathcal{M}_\lambda$
$X \,;\, Y$	composition = Boolean matrix product	≤ 2 of the $\mathcal{M}_\lambda$

every $\mathcal{M}_\lambda$ needs a composition that tells it apart $\implies$ $2^k \leq 2 \cdot \#\text{compositions}(C)$

スライド概要：C は $\mathcal{M}_0$ と 2^k 個の $\mathcal{M}_\lambda$ を見分けねばならない。合成 1 個が新たに見分けられるのは高々 2 個。

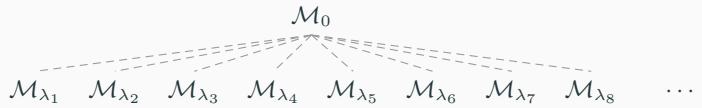
メモ：

- 「新たに見分ける」＝そのゲートで初めて $\mathcal{M}_0$ から期待される値とずれる（論文の notice）
- 以降のページ：変更（塗り替え）の作り方、期待値の意味、合成が高々 2 つの理由

Proof idea: telling $\mathcal{M}_0$ apart from 2^k structures

Goal $C \equiv G_k \implies C$ has $\geq 2^{k-1}$ compositions (C: any CoR term or circuit)

C must tell $\mathcal{M}_0$ apart from each of 2^k structures $\mathcal{M}_\lambda$: $G_k = 1$ on $\mathcal{M}_0$, $G_k = 0$ on $\mathcal{M}_\lambda$



CoR operation	meaning	newly tells $\mathcal{M}_0$ apart from
$\mathbf{0}, \mathbf{1}, \mathbf{id}$, $\cup, \cap, \bar{\cdot}$, $X^\smile$	constants, Boolean operations, converse	no $\mathcal{M}_\lambda$
$X \,;\, Y$	composition = Boolean matrix product	≤ 2 of the $\mathcal{M}_\lambda$

every $\mathcal{M}_\lambda$ needs a composition that tells it apart $\implies$ $2^k \leq 2 \cdot \#\text{compositions}(C)$

スライド概要：C は $\mathcal{M}_0$ と 2^k 個の $\mathcal{M}_\lambda$ を見分けねばならない。合成 1 個が新たに見分けられるのは高々 2 個。

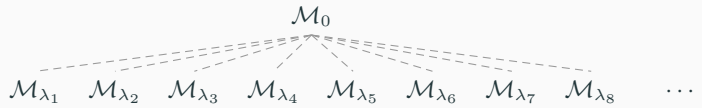
メモ：

- 「新たに見分ける」＝そのゲートで初めて $\mathcal{M}_0$ から期待される値とずれる（論文の notice）
- 以降のページ：変更（塗り替え）の作り方、期待値の意味、合成が高々 2 つの理由

Proof idea: telling $\mathcal{M}_0$ apart from 2^k structures

Goal $C \equiv G_k \implies C$ has $\geq 2^{k-1}$ compositions (C: any CoR term or circuit)

C must tell $\mathcal{M}_0$ apart from each of 2^k structures $\mathcal{M}_\lambda$: $G_k = 1$ on $\mathcal{M}_0$, $G_k = 0$ on $\mathcal{M}_\lambda$



CoR operation	meaning	newly tells $\mathcal{M}_0$ apart from
$\mathbf{0}, \mathbf{1}, \mathbf{id}$, $\cup, \cap, \neg$, $X^\smile$	constants, Boolean operations, converse	no $\mathcal{M}_\lambda$
$X \,;\, Y$	composition = Boolean matrix product	≤ 2 of the $\mathcal{M}_\lambda$

every $\mathcal{M}_\lambda$ needs a composition that tells it apart $\implies$ $2^k \leq 2 \cdot \#\text{compositions}(C)$

スライド概要：C は $\mathcal{M}_0$ と 2^k 個の $\mathcal{M}_\lambda$ を見分けねばならない。合成 1 個が新たに見分けられるのは高々 2 個。

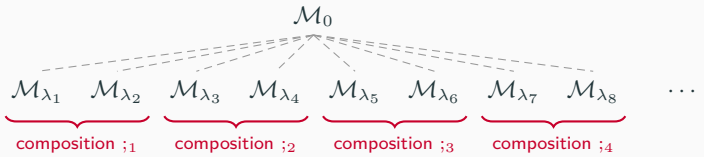
メモ：

- 「新たに見分ける」＝そのゲートで初めて $\mathcal{M}_0$ から期待される値とずれる（論文の notice）
- 以降のページ：変更（塗り替え）の作り方、期待値の意味、合成が高々 2 つの理由

Proof idea: telling $\mathcal{M}_0$ apart from 2^k structures

Goal $C \equiv G_k \implies C$ has $\geq 2^{k-1}$ compositions (C: any CoR term or circuit)

C must tell $\mathcal{M}_0$ apart from each of 2^k structures $\mathcal{M}_\lambda$: $G_k = 1$ on $\mathcal{M}_0$, $G_k = 0$ on $\mathcal{M}_\lambda$



CoR operation	meaning	newly tells $\mathcal{M}_0$ apart from
$\mathbf{0}, \mathbf{1}, \mathbf{id}$, $\cup, \cap, \bar{\cdot}$, $X^\smile$	constants, Boolean operations, converse	no $\mathcal{M}_\lambda$
$X \,;\, Y$	composition = Boolean matrix product	≤ 2 of the $\mathcal{M}_\lambda$

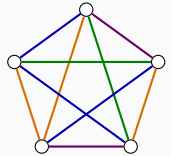
every $\mathcal{M}_\lambda$ needs a composition that tells it apart $\implies$ $2^k \leq 2 \cdot \#\text{compositions}(C)$

スライド概要：C は $\mathcal{M}_0$ と 2^k 個の $\mathcal{M}_\lambda$ を見分けねばならない。合成 1 個が新たに見分けられるのは高々 2 個。

メモ：

- 「新たに見分ける」＝そのゲートで初めて $\mathcal{M}_0$ から期待される値とずれる（論文の notice）
- 以降のページ：変更（塗り替え）の作り方、期待値の意味、合成が高々 2 つの理由

The structure $\mathcal{M}_0$



an edge-colored complete graph

colors $\mu \subseteq [k]$; on an edge of color μ: $P_i \iff i \in \mu$, $Q_i \iff i \notin \mu$

$\Rightarrow$ if xy has color μ: $G_k(x, y)$ holds iff some edge xz has color μ^c

for all $u \neq v$ and all colors μ, ν: at least **two** middle vertices z

(random coloring, $n \geq n_0(2^k)$ [JMT95, Algebra Univers.])

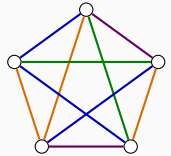
fix such an $\mathcal{M}_0$ and one vertex $\hbar$, the **hub**

スライド概要：$\mathcal{M}_0$ は辺に色の付いた完全グラフ。どの色の組にも中間の頂点が 2 つ。hub を 1 つ固定。

メモ：

- 色が P_i, Q_i を決めるので、G_k は色だけで読める
- 「中間の頂点が 2 つ」が最後のページで効く

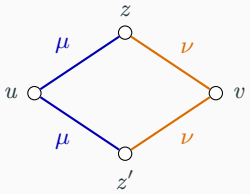
The structure $\mathcal{M}_0$



an edge-colored complete graph

colors $\mu \subseteq [k]$; on an edge of color μ: $P_i \iff i \in \mu$, $Q_i \iff i \notin \mu$

$\Rightarrow$ if xy has color μ: $G_k(x, y)$ holds iff some edge xz has color μ^c



for all $u \neq v$ and all colors μ, ν: at least **two** middle vertices z

(random coloring, $n \geq n_0(2^k)$ [JMT95, Algebra Univers.])

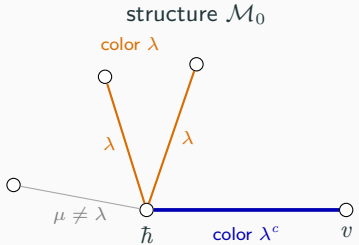
fix such an $\mathcal{M}_0$ and one vertex $\hbar$, the **hub**

スライド概要：$\mathcal{M}_0$ は辺に色の付いた完全グラフ。どの色の組にも中間の頂点が 2 つ。hub を 1 つ固定。

メモ：

- 色が P_i, Q_i を決めるので、G_k は色だけで読める
- 「中間の頂点が 2 つ」が最後のページで効く

Recoloring at the hub $\hbar$



$G_k(\hbar, v) = 1$

$G_k(\hbar, v)$ needs an edge $\hbar z$ of color λ

recoloring λ changes **exactly those edges at** $\hbar$, and nothing else

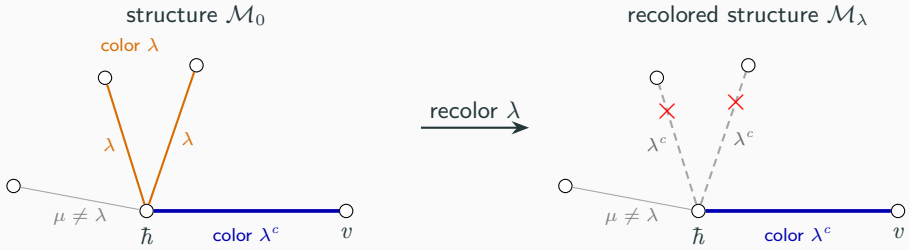
2^k recolored structures $\mathcal{M}_\lambda$, one for each color λ

スライド概要：塗り替えで G_k が 1 から 0 に変わる。

メモ：

- $\hbar$ に接する色 λ の辺だけ塗り替える
- 色ごとに 1 つ、計 2^k 個

Recoloring at the hub $\hbar$



$G_k(\hbar, v) = 1$ $\qquad$ $G_k(\hbar, v) = 0$

$G_k(\hbar, v)$ needs an edge $\hbar z$ of color λ
recoloring λ changes **exactly those edges at** $\hbar$, and nothing else

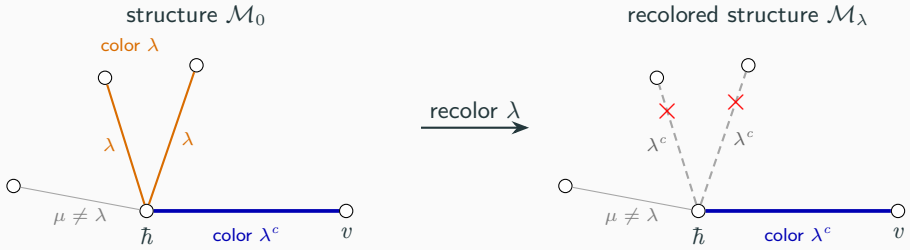
2^k recolored structures $\mathcal{M}_\lambda$, one for each color λ

スライド概要：塗り替えで G_k が 1 から 0 に変わる。

メモ：

- $\hbar$ に接する色 λ の辺だけ塗り替える
- 色ごとに 1 つ、計 2^k 個

Recoloring at the hub $\hbar$



$G_k(\hbar, v) = 1$ $\qquad$ $G_k(\hbar, v) = 0$

$G_k(\hbar, v)$ needs an edge $\hbar z$ of color λ
recoloring λ changes **exactly those edges at** $\hbar$, and nothing else

2^k recolored structures $\mathcal{M}_\lambda$, one for each color λ

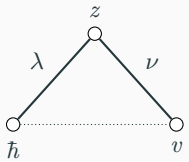
スライド概要：塗り替えで G_k が 1 から 0 に変わる。

メモ：

- $\hbar$ に接する色 λ の辺だけ塗り替える
- 色ごとに 1 つ、計 2^k 個

Why a composition detects at most two recolorings

(a) $\mathcal{M}_0$, at $(\hbar, v)$



middle vertex z:
$(X\,;\,Y)(\hbar, v) = 1$

a composition $X\,;\,Y$, where X holds on the edges of color λ and Y on those of color $\nu \neq \lambda$

only pairs at the hub can change, and only if X or Y singles out the color λ
$\Rightarrow$ each composition detects at most two colors λ

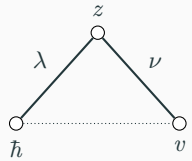
スライド概要：$\hbar$ を含む点対では気づかれうる、含まない点対では気づかれない。

メモ：

- $\hbar$ を含まない点対では、$\hbar$ を通らない中間の頂点 z' が残る
- 条件は 3 つ、どれも λ を決める $\rightarrow$ 高々 2

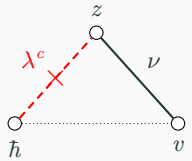
Why a composition detects at most two recolorings

(a) $\mathcal{M}_0$, at $(\hbar, v)$



middle vertex z:
$(X\,;\,Y)(\hbar, v) = 1$

(b) $\mathcal{M}_\lambda$, at $(\hbar, v)$



every middle vertex is lost:
detected

a composition $X\,;\,Y$, where X holds on the edges of color λ and Y on those of color $\nu \neq \lambda$

only pairs at the hub can change, and only if X or Y singles out the color λ
$\Rightarrow$ each composition detects at most two colors λ

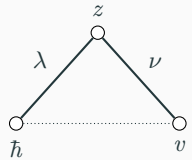
スライド概要：$\hbar$ を含む点対では気づかれうる、含まない点対では気づかれない。

メモ：

- $\hbar$ を含まない点対では、$\hbar$ を通らない中間の頂点 z' が残る
- 条件は 3 つ、どれも λ を決める $\rightarrow$ 高々 2

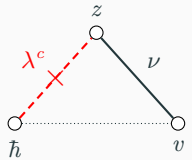
Why a composition detects at most two recolorings

(a) $\mathcal{M}_0$, at $(\hbar, v)$



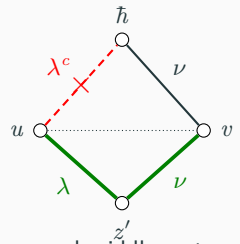
middle vertex z:
$(X\,;\,Y)(\hbar, v) = 1$

(b) $\mathcal{M}_\lambda$, at $(\hbar, v)$



every middle vertex is lost:
detected

(c) $\mathcal{M}_\lambda$, at (u, v)



a second middle vertex z'
avoids $\hbar$: not detected

a composition $X\,;\,Y$, where X holds on the edges of color λ and Y on those of color $\nu \neq \lambda$

only pairs at the hub can change, and only if X or Y singles out the color λ
$\Rightarrow$ each composition detects at most two colors λ

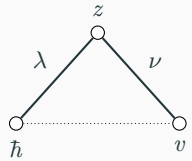
スライド概要：$\hbar$ を含む点対では気づかれうる、含まない点対では気づかれない。

メモ：

- $\hbar$ を含まない点対では、$\hbar$ を通らない中間の頂点 z' が残る
- 条件は 3 つ、どれも λ を決める $\rightarrow$ 高々 2

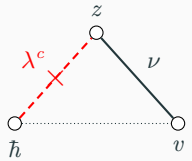
Why a composition detects at most two recolorings

(a) $\mathcal{M}_0$, at $(\hbar, v)$



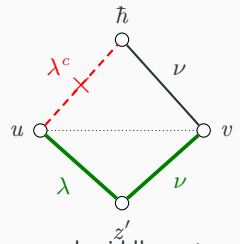
middle vertex z:
$(X\,;\,Y)(\hbar, v) = 1$

(b) $\mathcal{M}_\lambda$, at $(\hbar, v)$



every middle vertex is lost:
detected

(c) $\mathcal{M}_\lambda$, at (u, v)



a second middle vertex z'
avoids $\hbar$: not detected

a composition $X\,;\,Y$, where X holds on the edges of color λ and Y on those of color $\nu \neq \lambda$

only pairs at the hub can change, and only if X or Y singles out the color λ
$\Rightarrow$ each composition detects at most two colors λ

スライド概要：$\hbar$ を含む点対では気づかれうる、含まない点対では気づかれない。

メモ：

- $\hbar$ を含まない点対では、$\hbar$ を通らない中間の頂点 z' が残る
- 条件は 3 つ、どれも λ を決める $\rightarrow$ 高々 2

Outline

1. Our results

2. Technical overview

3. Conclusion

Summary

Result Tarski's translation from FO^3 to CoR is optimal in the worst case:

- some FO^3 formulas of size $O(k)$ need $\geq 2^{k-1}$ compositions in every equivalent CoR term, even with complement, converse, and shared subterms;
- the same technique: FO^3 vs. FO^2, matrix query languages, CoR with closure.

Technique 2^k recolorings of one structure $\mathcal{M}_0$; each composition detects ≤ 2 of them.

? Open questions

- a superpolynomial gap over a **fixed** finite signature (e.g., one binary relation)?
- the domain size from which the blowup begins?
- the exact number of compositions for G_k: between 2^{k-1} and 2^k?

Thank you for your attention!

スライド概要：結果・手法・未解決問題をまとめて締める。

メモ：

- 未解決：固定シグネチャ、開始の領域サイズ、正確な合成数

Summary

Result Tarski's translation from FO^3 to CoR is optimal in the worst case:

- some FO^3 formulas of size $O(k)$ need $\geq 2^{k-1}$ compositions in every equivalent CoR term, even with complement, converse, and shared subterms;
- the same technique: FO^3 vs. FO^2, matrix query languages, CoR with closure.

Technique 2^k recolorings of one structure $\mathcal{M}_0$; each composition detects ≤ 2 of them.

? Open questions

- a superpolynomial gap over a **fixed** finite signature (e.g., one binary relation)?
- the domain size from which the blowup begins?
- the exact number of compositions for G_k: between 2^{k-1} and 2^k?

Thank you for your attention!

スライド概要：結果・手法・未解決問題をまとめて締める。

メモ：

- 未解決：固定シグネチャ、開始の領域サイズ、正確な合成数

References i

[Var95] M. Y. Vardi. "On the Complexity of Bounded-Variable Queries". In: **Proceedings of the Fourteenth ACM SIGACT-SIGMOD-SIGART Symposium on Principles of Database Systems (PODS 1995)**. ACM, 1995, pp. 266–276.

[BS21] K. Bringmann and J. Slusallek. "Current Algorithms for Detecting Subgraphs of Bounded Treewidth Are Probably Optimal". In: **48th International Colloquium on Automata, Languages, and Programming (ICALP)**. Vol. 198. Leibniz International Proceedings in Informatics (LIPIcs). Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2021, 40:1–40:16. DOI: 10.4230/LIPIcs.ICALP.2021.40.

[NU27] Y. Nakamura and Y. Uezato. "A Fine-Grained Dichotomy for Bounded-Variable Query Evaluation: The Calculus of Relations, a Boolean Modal Logic, and One-Variable Counting Logic". In: **Proceedings of the ACM on Management of Data** 4.5 (PODS) (2027), 295:1–295:27. DOI: 10.1145/3850130.

[Wil14] R. Williams. "Faster Decision of First-Order Graph Properties". In: **Joint Meeting of the Twenty-Third EACSL Annual Conference on Computer Science Logic and the Twenty-Ninth Annual ACM/IEEE Symposium on Logic in Computer Science (CSL-LICS)**. ACM, 2014, 80:1–80:6. DOI: 10.1145/2603088.2603121.

References ii

[Tar41] A. Tarski. "On the Calculus of Relations". In: **The Journal of Symbolic Logic** 6.3 (1941), pp. 73–89. DOI: 10.2307/2268577.

[TG87] A. Tarski and S. Givant. **A Formalization of Set Theory without Variables**. Vol. 41. Colloquium Publications. Providence, RI: American Mathematical Society, 1987. DOI: 10.1090/coll/041.

[Nak22] Y. Nakamura. "Expressive Power and Succinctness of the Positive Calculus of Binary Relations". In: **Journal of Logical and Algebraic Methods in Programming** 127 (2022), p. 100760. DOI: 10.1016/j.jlamp.2022.100760.

[FGL+15] G. H. L. Fletcher, M. Gyssens, D. Leinders, D. Surinx, J. Van den Bussche, D. Van Gucht, S. Vansummeren, and Y. Wu. "Relative Expressive Power of Navigational Querying on Graphs". In: **Information Sciences** 298 (2015), pp. 390–406. DOI: 10.1016/j.ins.2014.11.031.

[Nak20] Y. Nakamura. "Expressive Power and Succinctness of the Positive Calculus of Relations". In: **Relational and Algebraic Methods in Computer Science (RAMiCS)**. Vol. 12062. LNCS. Springer, 2020, pp. 204–220. DOI: 10.1007/978-3-030-43520-2_13.

References iii

[VS24] H. Vinall-Smeeth. "From Quantifier Depth to Quantifier Number: Separating Structures with k Variables". In: **Proceedings of the 39th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)**. Full version: arXiv:2311.15885. ACM, 2024, 71:1–71:14.

[GS05] M. Grohe and N. Schweikardt. "The Succinctness of First-Order Logic on Linear Orders". In: **Logical Methods in Computer Science** 1.1:6 (2005). Conference version in LICS 2004, pp. 1–25. DOI: 10.2168/LMCS-1(1:6)2005.

[BGV19] R. Brijder, M. Gyssens, and J. Van den Bussche. **On Matrices and K-Relations**. arXiv:1904.03934v1. 2019. arXiv: 1904.03934 [cs.DB]. URL: https://arxiv.org/abs/1904.03934v1.

[BGV22] R. Brijder, M. Gyssens, and J. Van den Bussche. "On Matrices and K-Relations". In: **Annals of Mathematics and Artificial Intelligence** 90.2–3 (2022), pp. 181–210. DOI: 10.1007/s10472-021-09760-4.

[JMT95] P. Jipsen, R. D. Maddux, and Z. Tuza. "Small Representations of the Relation Algebra $E_{n+1}(1, 2, 3)$". In: **Algebra Universalis** 33 (1995), pp. 136–139. DOI: 10.1007/BF01190770.